

INE3015 — Production Planning and Control

Midterm Study Pack · 20 August, 10:30 · Lecture Notes 2, 3, 4

What the announcement says. The exam covers **Lecture Notes 2, 3 and 4** — Demand Forecasting, Sales & Operations Planning, and Inventory Management. **Lecture Note 1 (Introduction / competitive strategy) is out of scope.** All necessary formulas will be provided, and you must bring a calculator.

What that means for how you study. Do not spend time memorising formulas — spend it on (1) knowing *which* formula applies, (2) getting the *units* consistent, and (3) the course-specific conventions below, which a formula sheet will not tell you.

The five conventions that cost marks. MSE is divided by $n - 1$ in this course · net demand subtracts starting inventory in period 1 and adds desired ending inventory in the last period · the constant-workforce level comes from the **maximum** of (cumulative net demand / cumulative workdays) · for an all-units discount you compare the largest feasible EOQ against **every breakpoint above it** · for a discrete single-period problem you round **up** to the next stocking level whose cumulative probability reaches the service level.

Part 1 — Which Method, and the Traps in Each

1.1 Forecasting: choosing the technique

Data pattern	Technique	Key relation	Trap
No trend, no season	Naive	$F_{t+1} = A_t$	Cheapest benchmark; beats nothing but costs nothing
Noisy, level	Moving average (n)	Average of n most recent	Larger n = smoother but slower to react
Level, weight recency	Weighted MA	Weights must sum to 1	Heaviest weight on the most recent period
Level, smooth	Exponential smoothing	$F_t = F_{t-1} + \alpha(A_{t-1} - F_{t-1})$	Lags behind a trend — always biased low in an uptrend
Trend present	Linear trend equation	$F_t = a + bt$	t is the period index, not the year
Trend + smoothing	Trend-adjusted (TAF)	$TAF_{t+1} = S_t + T_t$	Two smoothing constants; update S before T
Seasonal	Seasonal relatives	Deseasonalise, project, reseasonalise	Divide to remove, multiply to put back
Seasonal + trend	CMA for indexes	CMA then relatives then average	Simple-average method is invalid when trend exists
Driven by another variable	Regression	$y = a + bx$	Predictor must lead the variable of interest

Accuracy measures. Error $e = A - F$ (actual minus forecast).

$MAD = \text{Sum}|e| / n$ · **$MSE = \text{Sum } e^2 / (n - 1)$** · $MAPE = \text{Sum}(|e|/A \times 100) / n$

MAD is in the units of the data and is the least sensitive to a single large error. MSE punishes large errors hardest.

MAPE is the only one that is unit-free, so it is the one to use when comparing series of different magnitudes. **Note the $n - 1$ in MSE — that is this course's convention, and dividing by n is the most common wrong answer.**

1.2 Seasonal relatives by centered moving average

Odd number of seasons (e.g. 3): a single CMA(3) is already centered on a period. **Even number (e.g. 4 quarters):** CMA(4) falls between two periods, so you must take a further 2-period average, CMA(2), to center it. Then relative = actual / centered average. Average the relatives belonging to the same season. **The seasonal indexes must sum to the number of seasons** (3.00 for three seasons, 4.00 for four) — if they do not, normalise them by multiplying each by (number of seasons / actual sum). Use this as your arithmetic check.

Deseasonalise by dividing each observation by its index; fit or project the trend on the deseasonalised series; then **reseasonalise** by multiplying the projection by the index of the target period.

1.3 Aggregate planning: the two benchmark plans

Step	Constant workforce (level)	Zero inventory (chase)
Net demand	Period 1: forecast – starting inventory. Last period: forecast + desired ending inventory.	Same net demand column.
Driver	Take the maximum of (cumulative net demand / cumulative workdays) over all periods. That period is the critical month .	Work period by period; no cumulative logic.
Workforce	$W = \text{max ratio} / K$, rounded up . Constant for the whole horizon.	$W_t = \text{net demand}_t / (K \times n_t)$, rounded up, possibly adjusted down.

Step	Constant workforce (level)	Zero inventory (chase)
Production	$P_t = K \times W \times n_t$	$P_t = K \times W_t \times n_t$
Inventory	Ending inv = cumulative production – cumulative net demand. Never negative by construction.	Near zero every period by design.
Cost	Hiring/firing once at the start + holding cost on a large inventory.	Large hiring and firing costs + almost no holding cost.

K = aggregate units produced by one worker in one day = total units / (workers x days). **Why the maximum ratio?** Because the straight production line drawn from the origin must lie *completely above* the cumulative net demand curve; the period where the curve is steepest on average is the binding one. Using the last period instead (total / total days) produces stockouts — that is the classic error.

The LP model: 8T variables and 3T constraints. $W_t = W_{t-1} + H_t - F_t$, $I_t = I_{t-1} + P_t + S_t - D_t$, $P_t = K n_t W_t + O_t - U_t$. Round W_t up (conservative approach): always feasible, rarely optimal. $O_t U_t = 0$ and $H_t F_t = 0$ — you never run overtime and idle time in the same period, nor hire and fire in the same period.

1.4 Inventory: picking the right model

Situation	Model	What to compute
Order arrives all at once, no discount	Basic EOQ	$Q^* = \sqrt{2DS/H}$; $TC = (Q/2)H + (D/Q)S$
You produce while you consume	EPQ	$Q_p = \sqrt{2DS/H} \times \sqrt{p/(p-u)}$; $I_{\max} = (Q_p/p)(p-u)$
Price falls with order size, all units	All-units discount	$TC = (Q/2)H + (D/Q)S + PD$, compared at candidate quantities
Price falls only above a breakpoint	Incremental discount	Optimum is always at a realizable EOQ
Constant demand and lead time	ROP certain	$ROP = d \times LT$
Demand varies	ROP, demand risk	$ROP = d\text{-bar} \times LT + z \times \sigma_d \times \sqrt{LT}$
Lead time varies	ROP, lead-time risk	$ROP = d \times LT\text{-bar} + z \times d \times \sigma_{LT}$
Both vary	ROP, both	$ROP = d\text{-bar}LT\text{-bar} + z \times \sqrt{LT\text{-bar} \sigma_d^2 + d\text{-bar}^2 \sigma_{LT}^2}$
Order at fixed times	FOI	$Q = T - IP$; $T = d\text{-bar}(OI+LT) + z \sigma_d \sqrt{OI+LT}$
Perishable, one order only	Single period	$SL = C_s / (C_s + C_e)$

Six inventory traps.

- 1. Check EOQ by equality.** At the true EOQ, annual carrying cost equals annual ordering cost. If your two numbers are not equal, your Q is wrong. This is a free check and it takes five seconds.
- 2. H may be a percentage of price.** If $H = i \times P$, then H changes with each discount price, so each price band has its own EOQ. Compute them all.
- 3. All-units procedure.** Compute the EOQ for each price; an EOQ is *realizable* only if it falls inside that price's quantity range. Take the largest realizable EOQ and compare its total cost against the total cost at **every breakpoint above it**. The winner is usually at a breakpoint, not at an EOQ.
- 4. Incremental is different.** The optimum is *always* at a realizable EOQ, never at a breakpoint. Compare the total costs at all realizable EOQs and stop.
- 5. p and u in the same units.** In EPQ, if p is per day then u must be per day: $u = \text{annual demand} / \text{working days per year}$. And $I_{\max}$ is **not** Q_p — it is smaller, because you are consuming while you produce.
- 6. Read which quantity varies.** Demand only, lead time only, or both — three different ROP expressions. Under demand risk the lead time multiplies as $\sqrt{LT}$, not LT .

Fixed quantity versus fixed interval. A fixed-quantity (ROP) system is triggered by *quantity* and needs protection only over the lead time. A fixed-interval system is triggered by *time* and needs protection over the **order interval plus the lead time** — a longer protection window, so it carries more safety stock for the same service level. Under higher-than-normal demand the fixed-quantity system reacts by ordering *sooner*; the fixed-interval system reacts by ordering a *larger amount* at the scheduled time. FOI requires only periodic review, which is its practical advantage.

ABC and cycle counting. A items: 10–20 per cent of items, 60–70 per cent of annual dollar value. C items: 50–60 per cent of items, 10–15 per cent of value. Pareto rule of thumb: first 20 per cent of items give 80 per cent of value, next 30 per cent give 15 per cent, last 50 per cent give 5 per cent. Counting accuracy required: A ± 0.2 per cent, B ± 1 per cent, C ± 5 per cent. **Classification is by annual dollar value = annual demand $\times$ unit cost, not by unit cost and not by demand alone.**

Part 2 — 75 True / False Statements

#	Statement	Ans	Why
Demand Forecasting (Lecture Note 2)			
1	Forecasts assume the same underlying causal system that existed in the past will continue.	T	First stated feature of forecasts.
2	Forecasts for groups of items tend to be more accurate than forecasts for individual items.	T	Errors offset across items.
3	Forecast accuracy increases as the forecasting horizon lengthens.	F	It decreases .
4	A good forecast should be stated in meaningful units and put in writing.	T	Elements of a good forecast.
5	Forecasts are rarely perfect.	T	Random variation cannot be forecast.
6	The Delphi method is a quantitative technique.	F	It is judgmental/qualitative.
7	Consumer surveys and salesforce opinions are judgmental forecasts.	T	Both qualitative.
8	Irregular variation is caused by unusual, identifiable events.	T	Distinct from random variation.
9	Random variation can be forecast if enough data is collected.	F	It is residual and unforecastable.
10	The naive forecast uses the most recent actual value as the next forecast.	T	$F_{t+1} = A_t$.
11	A larger n in a moving average makes the forecast respond faster to changes.	F	It responds slower and smooths more.
12	In a weighted moving average the weights must sum to 1.00.	T	Otherwise the forecast is biased.
13	In exponential smoothing a larger alpha gives more weight to recent data.	T	Faster response, less smoothing.
14	Simple exponential smoothing tracks a trended series without bias.	F	It lags persistently behind the trend.
15	Trend-adjusted exponential smoothing uses two smoothing constants.	T	alpha for level, beta for trend.
16	Forecast error is defined as forecast minus actual.	F	$e = \text{actual} - \text{forecast}$.
17	MSE in this course is computed by dividing the sum of squared errors by $n - 1$.	T	Course convention — not n.
18	MAPE is expressed as a percentage and is therefore unit-free.	T	Best for comparing different series.
19	MSE penalises large errors more heavily than MAD does.	T	Because errors are squared.
20	In the multiplicative model, seasonality is a quantity added to the series average.	F	That is the additive model.
21	To deseasonalise a series you divide each value by its seasonal relative.	T	Multiply to put seasonality back.
22	The seasonal indexes must sum to the number of seasons.	T	4.00 for quarters, 3.00 for three seasons.
23	The simple average method for seasonal relatives should be avoided when a linear trend is present.	T	Stated explicitly on the slide.
24	With an even number of seasons, a single moving average is already centered.	F	A further 2-period average is needed.
25	A proactive approach to using forecasts seeks to actively influence demand.	T	Advertising, pricing, product changes.
Sales and Operations Planning (Lecture Note 3)			
26	Aggregate planning determines workforce levels and production levels over the planning horizon.	T	Definition.
27	Aggregate planning is a detailed, item-level schedule.	F	It works in aggregate units; MPS is detailed.
28	Aggregate units may be tons, gallons, dollars, or fictitious units.	T	All listed as valid.
29	Aggregate planning assumes demand is known.	T	Uncertainty is deliberately ignored.
30	Smoothing costs arise from changing the workforce size or the production level.	T	Hiring and firing.
31	The primary component of holding cost is the opportunity cost of investment.	T	Stated on the slide.
32	In computing net demand you add the starting inventory to the first period's forecast.	F	You subtract it.
33	Desired ending inventory is added to the last period's forecast.	T	So the horizon ends with stock on hand.

#	Statement	Ans	Why
34	The constant workforce level is set by dividing total net demand by total workdays.	F	Use the maximum cumulative ratio.
35	The critical month is the one with the highest cumulative net demand per cumulative workday.	T	It binds the constant plan.
36	K is the number of aggregate units produced by one worker in one day.	T	$K = \text{units} / (\text{workers} \times \text{days})$.
37	The zero inventory plan is also called the chase strategy.	T	Workforce chases demand.
38	The chase plan generally has higher hiring and firing costs than the level plan.	T	It changes the workforce every period.
39	The level plan generally has higher holding costs than the chase plan.	T	It builds inventory ahead of peaks.
40	A constant production level always means a constant number of workers each month.	F	Months differ in workdays.
41	The LP formulation of aggregate planning has 8T variables and 3T constraints.	T	$T = \text{length of the horizon}$.
42	In the LP model, overtime and idle time can both be positive in the same period.	F	$O_t U_t = 0$.
43	Rounding the LP workforce up is always feasible but rarely optimal.	T	The conservative approach.
44	Backlogging is modelled by splitting inventory into two non-negative variables.	T	$I = I^+ - I^-$.
Inventory Management (Lecture Note 4)			
45	Independent demand items are those ready to be sold or used.	T	As opposed to dependent demand.
46	Work in process is a type of inventory.	T	One of the six types.
47	ABC classification is based on annual dollar value, not unit cost alone.	T	Demand x unit cost.
48	A items typically represent 50 to 60 per cent of the number of items.	F	That is C items.
49	C items account for about 10 to 15 per cent of annual dollar value.	T	Many items, little value.
50	Cycle counting accuracy required for A items is tighter than for C items.	T	+/-0.2 vs +/-5 per cent.
51	The basic EOQ model assumes quantity discounts are available.	F	It assumes there are none .
52	At the EOQ, annual carrying cost equals annual ordering cost.	T	Use it to check your answer.
53	The total cost curve in the EOQ model is U-shaped and flat near the optimum.	T	Small errors in Q cost little.
54	Doubling the ordering cost doubles the EOQ.	F	EOQ rises by $\sqrt{2}$, about 41 per cent.
55	Adding the purchasing cost PD shifts the total cost curve but does not move the EOQ.	T	PD is independent of Q.
56	In the EPQ model the maximum inventory equals the run size.	F	It is smaller: $(Q_p/p)(p - u)$.
57	In EPQ, p and u must be expressed in the same time units.	T	Both per day or both per week.
58	The EPQ run size is larger than the corresponding EOQ.	T	Multiplied by $\sqrt{p/(p-u)} > 1$.
59	For an all-units discount the optimum is often at a price breakpoint.	T	Generally at a breakpoint.
60	For an incremental discount the optimum is always at a realizable EOQ.	T	Never at a breakpoint.
61	When holding cost is a fixed percentage of price, every price band has its own EOQ.	T	$H = iP$ changes with P.
62	Under certainty the reorder point equals demand rate times lead time.	T	$ROP = d \times LT$.
63	Safety stock is held to protect against variability in demand and/or lead time.	T	Definition.
64	Service level is the probability that demand will not exceed supply during lead time.	T	$SL = 100 \text{ per cent} - \text{stockout risk}$.
65	Raising the service level from 95 to 99 per cent raises safety stock proportionally.	F	z rises non-linearly; cost accelerates.
66	Under demand uncertainty the safety stock term grows with the square root of lead time.	T	$z \times \sigma_d \times \sqrt{LT}$.
67	A fixed-interval system needs protection over lead time only.	F	Over the order interval plus lead time.

#	Statement	Ans	Why
68	A fixed-interval system generally requires more safety stock than a fixed-quantity system.	T	Longer protection interval.
69	In a fixed-quantity system, higher than normal demand shortens the time between orders.	T	The ROP is reached sooner.
70	In the single-period model, shortage cost is the unrealised profit per unit.	T	$C_s = \text{revenue} - \text{cost}$.
71	Excess cost is the cost per unit minus the salvage value per unit.	T	$C_e = \text{cost} - \text{salvage}$.
72	In the single-period model the optimal service level is $C_e / (C_s + C_e)$.	F	It is $C_s / (C_s + C_e)$.
73	For a discrete demand distribution you round down to the stocking level below the service level.	F	Round up to the next level.
74	If salvage value rises, the optimal stocking quantity rises.	T	C_e falls, so SL rises.

Part 3 — 40 Multiple Choice Questions

#	Question and options	Ans	Why
1	Forecast error is defined as: A) Forecast minus actual B) Actual minus forecast C) Absolute forecast D) Squared forecast	B	$e = A - F$.
2	Which accuracy measure is unit-free and therefore best for comparing different series? A) MAD B) MSE C) MAPE D) Bias	C	It is a percentage.
3	In this course MSE is computed as: A) $\text{Sum } e^2 / n$ B) $\text{Sum } e^2 / (n - 1)$ C) $\text{Sum} e / n$ D) $\text{Sum} e / (n - 1)$	B	Course-specific convention.
4	Increasing n in a moving average: A) Speeds up the response B) Smooths more and responds slower C) Has no effect D) Removes seasonality	B	More terms, more averaging.
5	Simple exponential smoothing applied to a series with an upward trend will: A) Overshoot the actuals B) Persistently lag below the actuals C) Track them exactly D) Oscillate around them	B	Hence the trend-adjusted version.
6	To remove seasonality from a series you: A) Multiply by the seasonal relative B) Divide by the seasonal relative C) Subtract the trend D) Take a naive forecast	B	Multiply to reseasonalise.
7	Four quarterly seasonal indexes must sum to: A) 1.00 B) 4.00 C) 100 D) The number of years	B	Number of seasons; use it as a check.
8	The simple average method of computing seasonal relatives should not be used when: A) Demand is high B) A linear trend is present C) There are four seasons D) Data covers three years	B	Use centered moving averages instead.
9	With quarterly data, computing centered relatives requires: A) One CMA(4) B) A CMA(4) followed by a CMA(2) C) A CMA(3) D) No moving average at all	B	An even span must be re-centered.
10	The Delphi method is best described as: A) An iterative judgmental technique using a panel of experts B) A regression technique C) A moving average variant D) A seasonal decomposition method	A	Qualitative.
11	In aggregate planning, net demand for the first period equals: A) Forecast + starting inventory B) Forecast – starting inventory C) Forecast + ending inventory D) Forecast only	B	Existing stock reduces what you must make.
12	Desired ending inventory is handled by: A) Subtracting it from the last period B) Adding it to the last period's forecast C) Ignoring it D) Spreading it over all periods	B	So the horizon closes with the target stock.

#	Question and options	Ans	Why
13	The constant workforce with no stockouts is determined by: A) Total net demand / total workdays B) The maximum of cumulative net demand / cumulative workdays C) The last period's ratio D) The average of the ratios	B	The line must lie above the whole cumulative curve.
14	K in aggregate planning is: A) Total cost per worker B) Aggregate units produced by one worker in one day C) The number of workdays D) The holding cost rate	B	$K = \text{units} / (\text{workers} \times \text{days})$.
15	The zero inventory plan is another name for: A) The level plan B) The chase plan C) The LP solution D) The rolling horizon	B	Workforce chases demand each period.
16	Compared with the level plan, the chase plan typically has: A) Higher holding cost, lower hiring cost B) Higher hiring and firing cost, lower holding cost C) Lower cost on both D) Identical costs	B	That is the trade-off being illustrated.
17	A constant monthly production level does not translate into a constant workforce because: A) Workers quit B) Months contain different numbers of workdays C) K changes monthly D) Demand is seasonal	B	$\text{Production} = K \times W \times n_t$.
18	The LP formulation of the aggregate planning problem has: A) 3T variables and 8T constraints B) 8T variables and 3T constraints C) T variables and T constraints D) 8 variables and 3 constraints	B	T is the horizon length.
19	Which pair of LP variables can never both be positive in the same period? A) P_t and I_t B) O_t and U_t C) W_t and P_t D) I_t and D_t	B	Overtime and idle time are mutually exclusive; the same holds for H_t and F_t .
20	ABC classification ranks items by: A) Unit cost B) Annual demand in units C) Annual demand multiplied by unit cost D) Order frequency	C	Annual dollar value.
21	A items typically account for: A) 10 to 20 per cent of items and 60 to 70 per cent of value B) 50 to 60 per cent of items and 10 to 15 per cent of value C) Half the items and half the value D) All slow-moving items	A	C items are the second description.
22	At the economic order quantity: A) Holding cost is minimised B) Annual carrying cost equals annual ordering cost C) Purchase cost is minimised D) Total cost equals purchase cost	B	The fastest check on your answer.
23	If the ordering cost S doubles, the EOQ: A) Doubles B) Halves C) Increases by a factor of $\sqrt{2}$ D) Is unchanged	C	EOQ varies with $\sqrt{S}$.
24	Adding the purchase cost term PD to the total cost function: A) Moves the EOQ to the right B) Moves the EOQ to the left C) Shifts the curve upward without moving the EOQ D) Makes the curve linear	C	PD does not depend on Q.
25	In the EPQ model, maximum inventory is: A) Equal to the run size Q_p B) $(Q/p)(p-u)$, which is less than Q_p C) $Q_p/2$ D) Equal to the usage rate	B	You consume while producing.

#	Question and options	Ans	Why
26	Compared with the basic EOQ, the EPQ run size is: A) Smaller B) Larger C) Equal D) Undefined	B	The factor $\sqrt{p/(p-u)}$ exceeds 1.
27	Under an all-units discount schedule, the optimal order quantity is found by: A) Taking the EOQ at the lowest price B) Comparing the largest realizable EOQ with every breakpoint above it C) Always ordering at the largest breakpoint D) Comparing only the realizable EOQs	B	The last option is the rule for <i>incremental</i> discounts.
28	Under an incremental discount schedule, the optimum occurs: A) At a breakpoint B) Always at a realizable EOQ C) At the largest quantity D) At the smallest quantity	B	The cost function has no downward jumps.
29	When holding cost is 20 per cent of the unit price, each price band: A) Shares one common EOQ B) Has its own EOQ C) Has no EOQ D) Has an EOQ of zero	B	$H = iP$ varies with P .
30	Under demand uncertainty only, the reorder point is: A) $d \times LT$ B) $\bar{d} \times LT + z \sigma_d \sqrt{LT}$ C) $d \times LT - \bar{d} + z d \sigma_{LT}$ D) $T - IP$	B	Safety stock scales with $\sqrt{LT}$.
31	Under lead-time uncertainty only, the reorder point is: A) $\bar{d} \times LT + z \sigma_d \sqrt{LT}$ B) $\bar{d} \times LT - \bar{d} + z d \sigma_{LT}$ C) $d \times LT$ D) $Q = T - IP$	B	The daily demand multiplies the lead-time spread.
32	Service level is defined as: A) The fraction of orders placed on time B) The probability that demand does not exceed supply during lead time C) The inventory turnover ratio D) One minus the holding cost rate	B	$SL = 100 \text{ per cent} - \text{stockout risk}$.
33	A fixed-interval system must carry stockout protection for: A) The lead time only B) The order interval plus the lead time C) One order cycle only D) No protection is needed	B	Hence more safety stock than an ROP system.
34	In a fixed-quantity system, demand that is higher than normal results in: A) A larger order quantity B) A shorter time between orders C) A longer lead time D) A higher target level	B	The ROP is hit sooner; Q is fixed.
35	The amount to order in a fixed-order-interval system is: A) The EOQ B) The target level minus the inventory position C) The ROP minus safety stock D) The average demand	B	$Q = T - IP$.
36	In the single-period model, the optimal service level equals: A) $C_e / (C_s + C_e)$ B) $C_s / (C_s + C_e)$ C) C_s / C_e D) $1 - C_s$	B	Shortage cost in the numerator.
37	Shortage cost per unit in the single-period model is: A) The purchase cost B) Revenue per unit minus cost per unit C) Cost minus salvage value D) The salvage value	B	The unrealised profit.

#	Question and options	Ans	Why
38	With a uniform demand distribution the optimal stocking level is: A) The midpoint of the range B) Minimum + SL x (maximum – minimum) C) The maximum D) The mean plus one standard deviation	B	SL is a proportion of the range.
39	With a discrete demand distribution you stock: A) The level whose cumulative probability is closest to SL B) The next higher level whose cumulative probability reaches at least SL C) The most likely demand D) The expected demand	B	Round up, never down.
40	If the salvage value of unsold units increases, the optimal stocking quantity: A) Falls B) Rises C) Is unchanged D) Becomes zero	B	C_e falls, so $SL = C_s / (C_s + C_e)$ rises.

Part 4 — 14 Calculator Problems with Full Solutions

These follow the lecture examples closely. Work them with the calculator you will bring; the point is the keystroke sequence and the unit bookkeeping, not the algebra.

Problem 1 — Forecast accuracy: MAD, MSE, MAPE

Actuals 217, 213, 216, 210, 213 with forecasts 215, 216, 215, 214, 211. Compute MAD, MSE and MAPE.

Solution. Errors $e = A - F$: +2, -3, +1, -4, +2

MAD = $(2+3+1+4+2)/5 = 12/5 = 2.40$

Sum $e^2 = 4+9+1+16+4 = 34 \rightarrow$ **MSE = $34/(5-1) = 8.50$** (dividing by 5 would give 6.80 — wrong here)

MAPE = $(2/217 + 3/213 + 1/216 + 4/210 + 2/213) \times 100 / 5 = (0.922+1.408+0.463+1.905+0.939)/5 = 1.13$ per cent

Problem 2 — Exponential smoothing

$\alpha = 0.10$. The forecast for period 5 was 213 and the actual turned out to be 216. Forecast period 6, then period 7 given an actual of 210.

Solution. $F_6 = 213 + 0.10(216 - 213) = 213 + 0.3 = 213.30$

$F_7 = 213.30 + 0.10(210 - 213.30) = 213.30 - 0.33 = 212.97$

Each new forecast is the old one nudged by alpha times the last error. Keep two decimals through the chain or the drift accumulates.

Problem 3 — Linear trend equation

Periods $t = 1..5$ with demands 42, 46, 49, 54, 57. Fit $F_t = a + bt$ and forecast period 8.

Solution. $n = 5$, Sum $t = 15$, Sum $t^2 = 55$, Sum $y = 248$, Sum $ty = 42+92+147+216+285 = 782$

$b = (5 \times 782 - 15 \times 248) / (5 \times 55 - 15^2) = (3910 - 3720) / (275 - 225) = 190/50 = 3.80$

$a = (248 - 3.80 \times 15) / 5 = (248 - 57) / 5 = 38.20$

$F_8 = 38.20 + 3.80(8) = 68.60$

Problem 4 — Seasonal relatives by centered moving average

Quarterly sales 40, 46, 42, 38, 44, 49, 47, 42, 47, 54, 51, 45. Find the four seasonal indexes.

Solution. CMA(4) then CMA(2) gives centered averages 42.00, 42.88, 43.88, 45.00, 45.88, 46.88, 48.00, 48.88 for quarters 3 through 10.

Relatives = actual / centered average: 1.00, 0.89, 1.00, 1.09, 1.02, 0.90, 0.98, 1.10

Average by season: $Q1 = (1.00+0.98)/2 = 0.99$ · $Q2 = (1.09+1.10)/2 = 1.10$ · $Q3 = (1.00+1.02)/2 = 1.01$ · $Q4 = (0.89+0.90)/2 = 0.90$

Check: $0.99 + 1.10 + 1.01 + 0.90 = 4.00$ — correct for four seasons.

Problem 5 — Deseasonalise, project, reseasonalise

Using the indexes above, period 12 demand is 45 (Q4) and the deseasonalised values for periods 11 and 12 are 50.50 and 50.56. Forecast period 13 (a Q1 period) with the naive trend method.

Solution. Deseasonalised naive with trend: $F_{13}^{(deseasonalised)} = 50.56 + (50.56 - 50.50) = 50.62$

Reseasonalise with the Q1 index: $F_{13} = 50.62 \times 0.99 = 50.11$

Divide to strip seasonality, project on the clean series, then multiply to put it back. Never project on the raw series.

Problem 6 — Aggregate planning: net demand and the critical month

Forecasts Jan–Aug: 420, 280, 460, 190, 310, 145, 110, 125. Starting inventory 200, desired ending inventory 100. Workdays 22, 16, 23, 20, 21, 17, 18, 10. Find net demand, the critical month, and the required constant workforce given $K = 0.3421$.

Solution. Net demand: $420-200 = 220$, then 280, 460, 190, 310, 145, 110, and $125+100 = 225$

Cumulative: 220, 500, 960, 1150, 1460, 1605, 1715, 1940

Cumulative workdays: 22, 38, 61, 81, 102, 119, 137, 147

Ratios: 10.00, 13.16, **15.74**, 14.20, 14.31, 13.49, 12.52, 13.20

Maximum is in March: $960/61 = 15.7377$ units per day $\rightarrow$ critical month = **March**

Workers = $15.7377 / 0.3421 = 46$

Problem 7 — Cost of the constant workforce plan

With 46 workers the monthly productions are 347, 252, 362, 315, 331, 268, 284, 158 and the ending inventories are 127, 99, 1, 126, 147, 270, 444, 477. The December workforce was 40. Holding cost \$8.50 per unit per month, hiring \$800 per worker.

Solution. Workers hired = $46 - 40 = 6 \rightarrow$ hiring cost = $6 \times 800 = \$4,800$

Total inventory-months = $127+99+1+126+147+270+444+477 = 1,691$

Holding cost = $1,691 \times 8.50 = \$14,373.50$

Total = \$18,973.50

Problem 8 — Cost of the chase (zero inventory) plan

The same problem solved by chasing demand requires 93 hires and 67 firings in total, and leaves 257 unit-months of inventory. Hiring \$800, firing \$1,250, holding \$8.50.

Solution. Hiring = $93 \times 800 = \$74,400$

Firing = $67 \times 1,250 = \$83,750$

Holding = $257 \times 8.50 = \$2,184.50$

Total = \$160,334.50

Roughly eight times the level plan: with hiring and firing this expensive relative to holding, chasing demand is the wrong strategy here. Always compare, never assume.

Problem 9 — Basic EOQ with all four sub-questions

A distributor expects to sell 9,600 tyres next year. Carrying cost is \$16 per tyre per year, ordering cost is \$75, and the firm operates 288 days a year. Find the EOQ, the number of orders, the cycle length, and the total annual cost.

Solution. $Q^* = \sqrt{2 \times 9,600 \times 75 / 16} = \sqrt{90,000} = 300$ tyres

Orders per year = $9,600/300 = 32$

Cycle = $300/9,600 = 1/32$ of a year = $288/32 = 9$ workdays

TC = $(300/2)16 + (9,600/300)75 = 2,400 + 2,400 = \$4,800$

The two components are equal — the answer checks out.

Problem 10 — EOQ when holding cost is a percentage of price

3,600 cameras a year at \$180 each, ordering cost \$50, carrying cost 20 per cent of purchase price.

Solution. $H = 0.20 \times 180 = \$36$

$Q^* = \sqrt{2 \times 3,600 \times 50 / 36} = \sqrt{10,000} = 100$ cameras

TC = $(100/2)36 + (3,600/100)50 = 1,800 + 1,800 = \$3,600$

Problem 11 — Economic production quantity

48,000 wheels a year, produced at 800 per day, carrying cost \$1 per wheel per year, setup cost \$45, 240 working days. Find the run size, the minimum total cost, the cycle time and the run time.

Solution. $u = 48,000/240 = 200$ per day (same units as p)

$Q_p = \sqrt{2 \times 48,000 \times 45 / 1} \times \sqrt{800/(800-200)} = 2,078.5 \times 1.1547 = 2,400$ wheels

$I_{\max} = (2,400/800)(800 - 200) = 1,800$ wheels — not 2,400

TC = $(1,800/2)(1) + (48,000/2,400)(45) = 900 + 900 = \$1,800$

Cycle time = $2,400/200 = 12$ days · Run time = $2,400/800 = 3$ days

Problem 12 — All-units quantity discount

816 cases a year, ordering cost \$12, carrying cost \$4 per case per year. Prices: \$20 for 1–49, \$18 for 50–79, \$17 for 80–99, \$16 for 100 or more.

Solution. H does not depend on price here, so there is one common EOQ:

$Q = \sqrt{2 \times 816 \times 12 / 4} = 69.97 \approx 70$ cases, which falls in the \$18 band, so it is realizable.

Compare 70 with every breakpoint above it:

TC(70) = $(70/2)4 + (816/70)12 + 18(816) = 140 + 139.9 + 14,688 = \$14,968$

TC(80) = $(80/2)4 + (816/80)12 + 17(816) = 160 + 122.4 + 13,872 = \$14,154$

TC(100) = $(100/2)4 + (816/100)12 + 16(816) = 200 + 97.9 + 13,056 = \$13,354$

Order 100 cases. The purchase term dominates, which is why the optimum sits at a breakpoint.

Problem 13 — All-units discount when H depends on price

600 units a year, ordering cost \$8, holding rate 20 per cent a year. Price is \$0.30 below 500 units, \$0.29 for 500–999, \$0.28 for 1,000 and above.

Solution. Each band has its own EOQ because $H = 0.20P$:

$Q(0.30) = \sqrt{2 \times 600 \times 8 / (0.20 \times 0.30)} = 400$ — inside 0–499, realizable

$Q(0.29) = 406$ — not inside 500–999, not realizable

$Q(0.28) = 414$ — not inside 1,000+, not realizable

Compare the realizable EOQ with the breakpoints above it:

TC(400) = $400/2(0.20 \times 0.30) + (600/400)8 + 600(0.30) = 204.00$

TC(500) = $500/2(0.20 \times 0.29) + (600/500)8 + 600(0.29) = 198.10$

TC(1000) = $1000/2(0.20 \times 0.28) + (600/1000)8 + 600(0.28) = 200.80$

Order 500 units at \$198.10 a year.

Problem 14 — Incremental quantity discount

Same data as above, but the discount applies only to units above each breakpoint.

Solution. $P(Q) = 0.30Q$ for $Q < 500$ · $150 + 0.29(Q-500) = 5 + 0.29Q$ for $500 \leq Q < 1000$ · $295 + 0.28(Q-1000) = 15 + 0.28Q$ for $Q \geq 1000$

Minimising each band's total cost gives:

$Q_1 = \sqrt{2 \times 600 \times 8 / (0.20 \times 0.30)} = 400$ (realizable) · $Q_2 = \sqrt{2 \times 13 \times 600 / (0.20 \times 0.29)} = 519$ (realizable) · $Q_3 = 702$ (not realizable)

TC(400) = 204.00 vs TC(519) = 204.58

Order 400 units. Note the optimum is at an EOQ, never at a breakpoint — the opposite of the all-units rule.

Problem 15 — Reorder point under each kind of uncertainty

Average daily demand 30 units, average lead time 4 days, sigma of daily demand 5 units, sigma of lead time 0.5 days, $z = 1.65$ for a 95 per cent service level.

Solution. Demand varies only: $ROP = 30(4) + 1.65(5)\sqrt{4} = 120 + 16.5 = 136.5 \approx 137$

Lead time varies only: $ROP = 30(4) + 1.65(30)(0.5) = 120 + 24.75 = 144.75 \approx 145$

Both vary: $ROP = 120 + 1.65 \times \sqrt{4(25) + 900(0.25)} = 120 + 1.65 \times \sqrt{325} = 120 + 1.65(18.03) = 120 + 29.7 = 149.7 \approx 150$

Safety stock is 16.5, 24.75 and 29.7 respectively. Read the question carefully — picking the wrong one of the three is the single most common inventory mistake.

Problem 16 — Single-period model, three distributions

(a) Cider: cost \$0.20 per litre, price \$0.80, no salvage, demand uniform between 300 and 500 litres. (b) A blend with the same costs but normally distributed demand, mean 200, sigma 10. (c) Spare parts: shortage cost \$4,200, unit cost \$800, zero salvage, demand 0/1/2/3 with probabilities 0.20/0.40/0.30/0.10.

Solution. $C_s = 0.80 - 0.20 = 0.60$ · $C_e = 0.20 - 0 = 0.20 \rightarrow SL = 0.60/(0.60+0.20) = 0.75$

(a) Uniform: $S_o = 300 + 0.75(500 - 300) = 450$ litres; stockout risk = $1 - 0.75 = 0.25$

(b) Normal: $SL = 0.75 \rightarrow z = 0.67 \rightarrow S_o = 200 + 0.67(10) = 206.7$ litres

(c) $SL = 4,200/(4,200+800) = 0.84$. Cumulative probabilities are 0.20, 0.60, 0.90, 1.00. The first level reaching 0.84 is **2 spares** — round up, never to the nearest.

Part 5 — Exam Tactics and Self-Check List

5.1 Before you compute anything

- 1. Identify the model in one line.** Write down which one you are using before touching the calculator: it stops you halfway into the wrong formula.
- 2. Fix the time unit.** Annual demand, daily usage, weekly lead time — convert everything to one base first, and write the conversion down. Most lost marks in this course are unit errors, not algebra errors.
- 3. Write the given quantities as symbols.** $D = \dots$, $S = \dots$, $H = \dots$, $p = \dots$, $u = \dots$. Then the formula sheet becomes a lookup rather than a puzzle.
- 4. Sanity-check the magnitude.** An EOQ larger than annual demand, a workforce of 4,000 for a plant of 300, a service level above 1 — all mean a slipped decimal.

5.2 The built-in checks

Topic	Free check
Seasonal indexes	They must sum to the number of seasons (4.00 for quarters).
EOQ	Carrying cost must equal ordering cost at Q^* .
EPQ	$I_{\max}$ must be less than Q_p ; run time must be less than cycle time.
Constant workforce	Ending inventory must be non-negative in every period, and smallest in the critical month.
Chase plan	Ending inventory should be near zero every period except possibly the last.
Aggregate planning	Cumulative production minus cumulative net demand equals ending inventory — recompute one row at random.
Service level	It must lie strictly between 0 and 1; if $C_s > C_e$ it must exceed 0.5.
ROP	Safety stock is the part beyond expected lead-time demand; it must be positive and modest relative to it.

The ten highest-value facts, if you only have ten minutes.

1. $e = A - F$; **MSE divides by $n - 1$** ; MAPE is the unit-free one.
2. Deseasonalise by **dividing** by the index, reseasonalise by **multiplying**; indexes sum to the number of seasons.
3. Simple exponential smoothing **lags** a trend; that is why trend-adjusted smoothing exists.
4. Net demand: **subtract** starting inventory in period 1, **add** desired ending inventory in the last.
5. Constant workforce comes from the **maximum** cumulative ratio, rounded up; that period is the critical month.
6. Level plan = high holding, low hiring. Chase plan = high hiring and firing, low holding. Compare, never assume.
7. At the EOQ, **carrying cost = ordering cost**. Adding PD does not move the EOQ.
8. All-units: largest realizable EOQ versus **all breakpoints above it**. Incremental: **realizable EOQs only**.
9. Three ROP formulas — demand risk uses $\sqrt{LT}$, lead-time risk uses $d \times \sigma_{LT}$, both combine under one square root.
10. Single period: $SL = C_s / (C_s + C_e)$; for discrete demand round **up** to the next cumulative level.